Assignment :- 4

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subject: TOC

① Explain pumping lemma for CFL. Prove that $L = \{a^n b^n c^n \mid n \geq 0\}$ is not CFL.

→ Let 'L' be any CFL then there is a constant 'n' which depends only upon L such that there exists a string $w \in L$ and $|w| \geq n$ then string 'w' can be divided into 5 sub strings p, q, r, s, t such that

conditions to be satisfied :-

(I) $|qs| > 0$

(II) $|qrs| \leq n$

(III) $\forall i \geq 0, \; pq^i r s^i t \in L$

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a) $L = \{a^n b^n c^n \mid n \geq 0\}$

Solution,

Assume the given language is PDA and let's assume a string for $n = 2$ then,

$\omega = \underset{p\ q}{aa}\ \underset{r\ s}{bb}\ \underset{T}{cc}$,

$\therefore |\omega| = 6 = n$

① $|qs| > 0 \quad \therefore 2 > 0$

② $|prs| \leq n \quad \therefore 3 \leq 6$

③ $\forall i \geq 0, \; pq^i r s^i T \in L$

for $i = 0$,

$\omega = a(a)^0 b(b)^0 cc$

$= abcc \notin L$

Hence, it is proved that our assumption is false so the above language is not a CFL and we cannot make a PDA for it.

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② Explain Turing Machine and working of Turing machine.

→ They are abstract computational devices used to explore the ~~limits~~ limits of what can be computed.
A Turing machine (TM) has an infinite tape, a read/write head, and rules that control how it reads, writes and moves on the tape. It can simulate any computation, making it as powerful as modern computers.

The machine starts in the initial state and follows transition rules until it reaches an accept or reject state.

In automata theory, TM are used to study algorithms, computability, and computational complexity.

The behaviour of TM is controlled by a finite state machine consisting of a finite set of states, a transition function and start and accept states.

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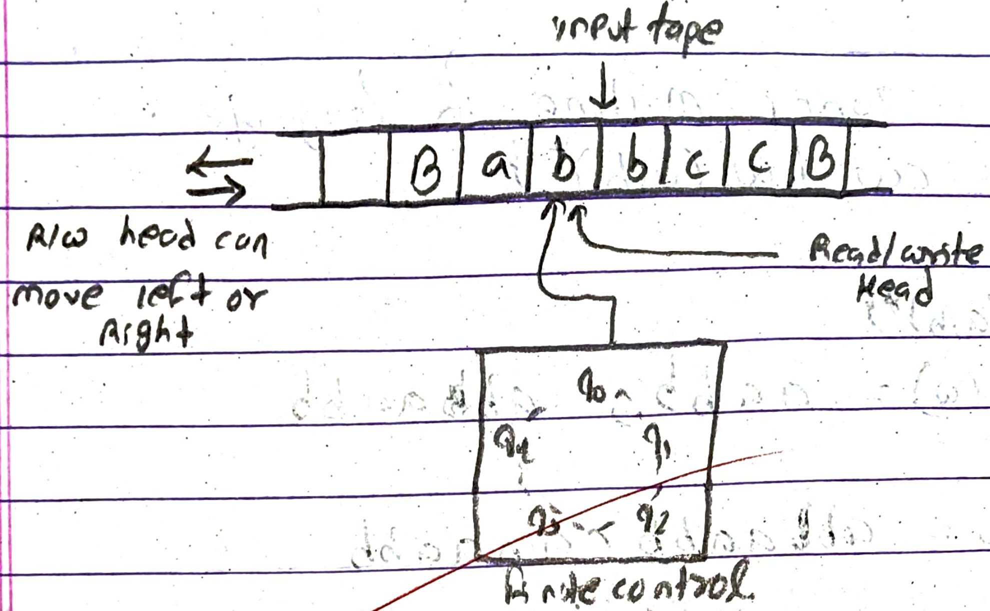

Turing machine formalism :-

1) A finite set of states (Q).
2) An input alphabet (Σ).
3) A tape alphabet (Γ) that includes Σ.
4) A transition function (δ)
5) A start state (q_0)
6) A blank symbol (B)
7) A set of final states (F)

$q_0 = q_0 \in Q$

$F = F \subseteq Q$

B = Blank

T = Tape symbol ($\Sigma \cup B, x \ldots$)

$\delta = (Q \times \Gamma \to Q \times \Gamma \times \{L/R\})$

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③ Design a Turing machine for language
$L = \{wcw \mid w\ (a,b)^*\}$

Solution,

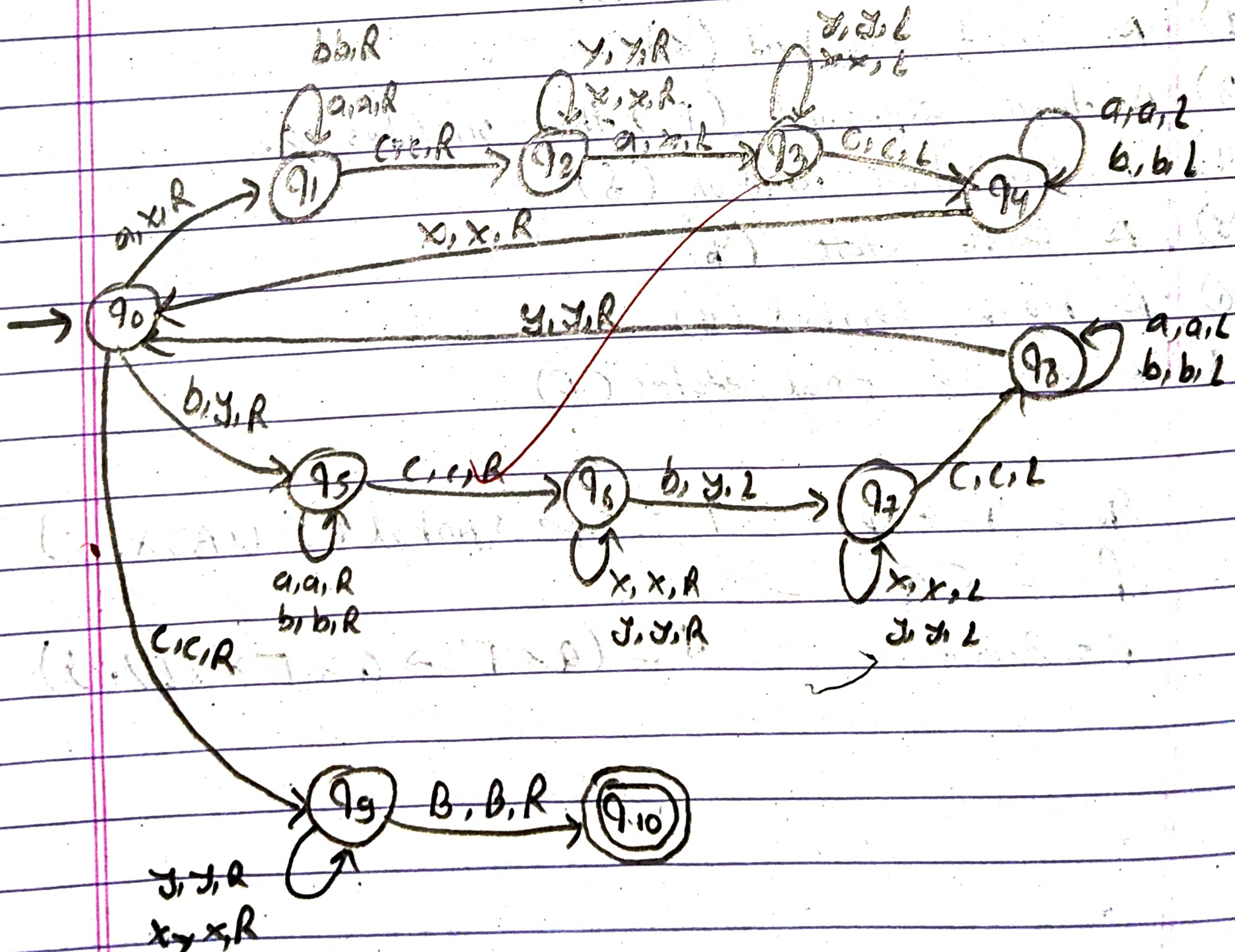
$\Sigma = \{(a,b)^*\}$

string (w) = aabb, ab~~b~~aabb

∴ wcw = ab~~b~~aabbcabaabb

	x	y	x	x	y	y		x	y	x	x	y	y	
B	a	b	a	a	b	b	c	a	b	a	a	b	b	B

Tape:



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Transition table.

	a	b	x	y	c	B
q_0	q_1, x, R	q_5, y, R			q_9, c, R	
q_1	q_1, a, R	q_1, b, R			q_2, c, R	
q_2	q_3, x, L		q_2, x, R	q_2, y, R		
q_3			q_3, x, L	q_3, y, L	q_4, c, L	
q_4	q_4, a, L	q_0, b, L	q_0, x, R			
q_5	q_5, a, R	q_5, b, R			q_6, c, R	
q_6		q_7, y, L	q_6, x, R	q_6, y, R		
q_7			q_7, x, L	q_7, y, L	q_8, c, L	
q_8	q_8, a, L	q_8, b, L		q_0, y, R		
q_9			q_9, x, R	q_9, y, R		q_{10}, B, R
q_{10}						

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(4) Design a Turing Machine for Language $L = \{ww^R \mid w \{a,b\}^*\}$

Solution,

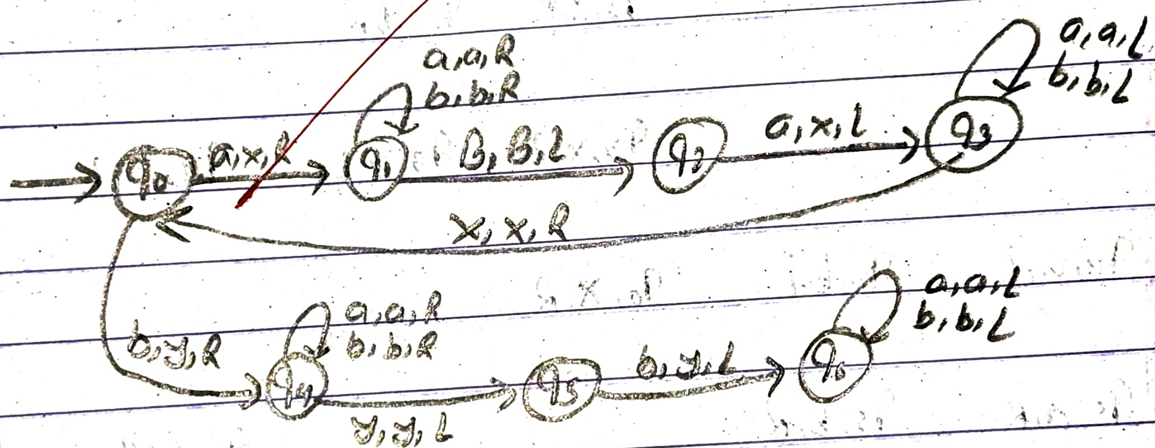
$\Sigma = \{a,b\}$, ww^R = aabb aa

$\therefore ww^R$ = aabbaa

a → x
y → b

Tape:

B	~~a~~	~~a~~	~~b~~	~~b~~	~~a~~	~~x~~	B



Transition table

	a	b	x	y	B
q_0	q_1,x,R	q_4,y,R			q_2,B,L
q_1	q_1,a,R	q_1,b,R			
q_2			q_3,x,L		
q_3	q_3,a,L	q_3,b,L	q_0,x,R		
q_4	q_4,a,R	q_4,b,R		q_5,y,L	
q_5		q_6,y,L			
q_6	q_6,a,L	q_6,b,L			

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⑤ Explain variants of Turing Machine

⇒ Variants of a Turing Machine are modified designed to make computation easier or more efficient. Even though their structure changes, all variants have the same computational power as a standard Turing machine.

Key Variants with Examples:

i) Multi-Tape TM:
uses more than one tape for Processing.
example: one tape stores input 1011, another tape is used to copy and reverse it for Palindrome Checking.

ii) Multi-Track TM:
A single tape divided into multiple tracks.
ex: Track 1 stores input abc, Track 2 stores markers like xx to track visited symbols.

iii) Non-Deterministic TM (NTM):
can take multiple paths at once.
ex: while checking if a string is a palindrome, it can guess the middle and verify both sides simultaneously.

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iv) Multi-Head TM
- multiple head read/write on the same tape.
ex: one head reads the start of a string and another reads the end to compare characters.

v) Two-Dimensional TM:
- Tape is like a grid instead of a line.
ex: used in problems like maze solving where movements is in all directions.

vi) off-line TM
- separate read-only input tape and working tape.
ex: input remains unchanged while computation happens on another tape.

vii) Linear Bounded Automata (LBA):
Tape size is limited to input length.
ex: used in parsing context-sensitive languages like a^b^c^

viii) Enumerator:
Generates strings instead of accepting/rejecting.
ex: Prints all strings of a language like {a, aa, aaa, ...}

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⑥ Explain Universal Turing Machine.

→ A universal Turing machine that can simulate any other Turing machine. It acts like a general-purpose system that can perform different tasks based on the input it receives.

- Takes input in the form ⟨M, W⟩
  - M = description of a machine
  - w = input string
- example: If M is a machine that checks whether a string has equal number of 0s and 1s, and w = 0011, then UTM simulates M on 0011 and gives the result.

- works like a computer running programs

ex:
Laptop = UTM
software (like a calculator app) = M
Numbers you enter = w
can simulate any algorithm or ~~comp~~ computation

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